

Analytics in the Octagon: Modeling UFC Fight Prediction

Group 4 Project Final Report

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ABSTRACT

This project explores whether historical UFC fight data can be effectively used to predict bout outcomes using machine learning techniques. The primary questions are: Can fighter attributes combined with derived comparative measures forecast the outcome of a fight, and how do these predictions compare with bookmaker odds? Using the “Ultimate UFC Dataset” from Kaggle, I engineered features such as reach, height, and age differences and applied three classifiers—Logistic Regression, Random Forest, and XGBoost—with hyperparameter tuning on all models.

My experiments show that ensemble methods, particularly Random Forest and a tuned version of XGBoost, capture the nonlinear dynamics inherent in fight data. The best models achieved approximately 64.5% accuracy and a 70.5% F1 score, offering actionable insights for fighters, coaches, and betting analysts.

INTRODUCTION

1.1 Background and Motivation

Mixed Martial Arts (MMA) is growing rapidly in popularity, and the one-on-one nature of UFC bouts makes them a compelling challenge for predictive analytics. Unlike team sports—where collective performance is key—MMA outcomes depend on individual physical attributes and the fight context. This project aims to determine whether machine learning can reliably forecast fight outcomes and generate insights that could inform training practices and betting strategies.

1.2 Research Questions

My key objectives are:

- **Predictive Modeling:** Can physical features (e.g., height, reach, age) along with differences derived from these measures predict fight outcomes accurately?
- **Model Comparison:** How do different models—Logistic Regression, Random Forest, and XGBoost (each tuned via GridSearchCV)—perform in terms of accuracy and robustness?
- **Benchmarking:** How do these model predictions compare with the implied probabilities from bookmaker odds?

1.3 Significance

Insights from this analysis can directly influence training methodologies, strategic game-planning, and even expose inefficiencies in betting markets. This report builds on my earlier work and details an enhanced modeling pipeline that now includes consistent hyperparameter tuning across all classifiers.

RELATED WORK

Most sports analytics research has traditionally focused on team sports. However, there is a growing body of work on predicting outcomes in individual sports like MMA. Prior studies have used decision trees and ensemble techniques to capture complex interactions among fighter attributes and performance metrics. My approach extends these ideas by integrating advanced feature engineering along with comprehensive hyperparameter tuning to ensure that each model is optimized for the nuances of the dataset.

DATA SET

3.1 Data Source

I used the “Ultimate UFC Dataset” from Kaggle (contributed by mdabbert), which includes data on over 6,500 UFC bouts. <https://www.kaggle.com/datasets/mdabbert/ultimate-ufc-dataset>

Essential data includes:

- **Fighter Attributes:** Height, reach, age, stance, etc.
- **Bout Details:** Date, location, and title bout status.
- **Outcome Data:** Winner (categorized as Red or Blue) along with betting odds.

3.2 Data Cleaning and Preprocessing

The data was prepared by:

- Dropping irrelevant columns such as fighter names, dates, and locations.
- Handling missing values by imputing the median for numeric fields and the mode for categorical ones.
- Encoding the “Winner” variable (Red as 1, Blue as 0) and label-encoding other categorical variables.
- Engineering new features such as *ReachDiff*, *HeightDiff*, and *AgeDiff* to capture competitive differences.
- Standardizing numeric features using StandardScaler.

MAIN TECHNIQUES APPLIED

4.1 Data Processing Pipeline

The pipeline efficiently converts raw data into a model-ready format by:

- **Cleaning:** Removing non-informative columns.
- **Imputation & Encoding:** Managing missing data.

- **Feature Engineering:** Creating metrics that quantify differences between fighters.
- **Scaling:** Normalizing features for improved model performance.

4.2 Modeling

I implemented three classifiers, each with hyperparameter tuning via GridSearchCV:

- **Logistic Regression:** Used as a baseline model and tuned over the regularization parameter C ([0.01, 0.1, 1, 10]).
- **Random Forest:** An ensemble method tuned for parameters such as n_estimators (fixed at 100), max_depth ([10, 20]), and min_samples_split (5).
- **XGBoost:** Initially set with default parameters, now updated to include hyperparameter tuning with a grid over:
 - n_estimators: [50, 100, 200]
 - max_depth: [3, 4, 5]
 - learning_rate: [0.01, 0.1, 0.2]
 - subsample: [0.8, 1.0]

GridSearchCV ensures all models are tuned consistently.

4.3 Evaluation

After an 80/20 train-test split and balanced subsampling to create a training set of 4,000 rows, I evaluated the models using metrics like accuracy, precision, recall, and F1 score. Random Forest emerged as the top performer with approximately 64.5% accuracy and a 70.5% F1 score, while the tuned XGBoost model provided competitive results, underscoring the benefit of thorough hyperparameter tuning.

KEY RESULTS

5.1 Quantitative Outcomes

- **Random Forest:** Approximately 64.5% accuracy and a 70.5% F1 score.

- Logistic Regression: Served as a reliable baseline.
- XGBoost (Tuned): Improved performance that brought its metrics closer to those of Random Forest, confirming the value of the tuning process.

5.2 Insights from Feature Engineering

Engineered features—specifically the differences in reach, height, and age—proved to be critical predictors. Larger disparities in these metrics often correlated with more decisive outcomes, supporting the hypothesis that physical differences significantly influence fight results.

5.3 Benchmarking Against Betting Odds

Preliminary results indicate that the tuned models may occasionally reveal discrepancies between predicted outcomes and bookmaker odds, hinting at potential market inefficiencies.

APPLICATIONS

6.1 Coaching and Training

The analysis can help coaches and fighters focus on key physical attributes that substantially impact performance.

6.2 Betting Analytics

Model predictions, when compared with bookmaker odds, can serve as a quantitative basis for identifying potential mispricing in the betting market.

6.3 Broadcasting Enhancements

Even without interactive visualizations, the derived insights can support richer, data-driven commentary during live events.

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REFERENCES

- [1] K. Williams. 2021. “Predicting MMA Fight Outcomes Using Machine Learning.” *Proceedings of 2021 Sports Analytics Conference*.
- [2] R. Hernandez. 2020. “Exploring Reach, Stance, and Takedown Defense in UFC Fighters.” *International Journal of Combat Sports Analytics*.
- [3] A. Smith, B. Johnson, and C. Lee. 2019. “Betting Markets and Probabilistic Forecasts: Evidence from the UFC.” *Journal of Sports Economics*.
- [4] David Kosiur. 2001. *Understanding Policy-Based Networking* (2nd. ed.). Wiley, New York, NY.